

AGENEWTECH

M328S

Smart Module Specification_V2.1



Our aim is to provide customers with timely and comprehensive service. For any assistance, please contact our company headquarters:

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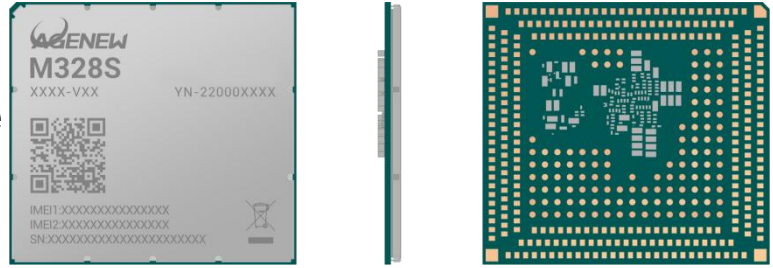
About the Document

History

Version	Date	Author	Description
-	2025-03-25	Daniel Wu	Initial
1.0	2025-03-27	Daniel Wu	Controlled
1.1	2025-07-01	Daniel Wu	Update the software version to Android15
1.2	2025-08-19	Xiaochen Huang	Update and supplement interface
2.0	2025-12-18	Daniel Wu	Disable the WIFI and Bluetooth functions
2.1	2026-02-09	Daniel Wu	Update the company name

M328S

Smart Computing Power Module



M328S is a smart computing module based on MediaTek's Octa-core MT8391 6nm processor (2×ARM Cortex-A78@2.6GHz + 6×ARM Cortex-A55@2.0GHz) launched by AGENEW, with inbuilt ARM Mali G57 MC2 GPU and NPU 8.0, equipped with Android 15 operating system. Its powerful performance, neon engine processing, 2D/3D graphics acceleration capabilities, fully meet the needs of customers for high-speed and high-computing-power functions in industrial and consumer applications.

M328S has audio and video input/output interfaces and rich GPIO interfaces, including LCM display modules, touchscreens, cameras, microphones, speakers, UART, USB, I2C, SPI, PCIe, etc. It supports various audio and video codecs. The main screen supports up to UW5K60(5120×2160) @60fps or 4K60(3840×2160) @60fps. AI computing power reaches 10 TOPS.

M328S is a SMT module, with a total of 455 pins, all of which are LGA pins, with a size of 42mm×46mm×2.6mm, which can be embedded in various M2M product applications through the pad, making it widely used in video conferencing, live streaming, gaming, edge computing, robotics, drones, augmented reality/virtual reality (AR/VR), and other terminal products and industries.



Main Advantages

- ✓ Built-in AI unit, support 10T computing power
- ✓ Support Android 15
- ✓ Support multiple audio and video codecs
- ✓ Main screen supports up to UW5K60(5120×2160) @60fps or 4K60(3840×2160) @60fps
- ✓ Support DP1.4
- ✓ Support 4K@60fps video recording and playback
- ✓ Abundant audio/video input and output interfaces and GPIO interfaces



64-bit Processor (Octa-core)



Support Android 15



10 TOPS



Powerful Multimedia

AI	M328S
Region/Operator	China
Operating System	Android 15
Memory	128GB UFS 3.1 + 8GB LPDDR5X
Module Dimension (mm)	42 × 46 × 2.6
Packaging	LGA
Temperature Range	
Operation Temperature	Operating temperature range: -20 °C ~ + 80 °C Storage temperature range: -40 °C ~ + 90 °C
Certification	
Compulsory Certification	Undetermined
Other Certifications	Undetermined
Features	
AI	10 TOPS
Video Encoding	4K@30fps
Video Decoding	4K@60fps
USB Firmware Update	•
OTA Firmware Update	•
Electrical Specification	
Operating Voltage	3.4~4.5 V, typical 3.8 V
Consumption (Typical)	200μA @ power off, 5.5mA @ sleep mode, typical
Interface	
LCD	Support up to UW5K60(5120×2160)@60fps or 4K60(3840×2160)@60fps 2 × 4-lane MIPI DSI,eDP1.4 4-lane, DP1.4 4-lane (Type-C)
Camera	4-lane MIPI CSI, Support Single camera up to 32MP@30fps Support double cameras (16MP + 16MP)@30fps
TP	Capacitive touch screen, I2C control
Audio	Lineout Interface (Require external audio amplifier) × 1, headphone × 1, simulated MIC ×4, digital MIC ×2
USB	USB2.0 × 3, USB 3.2 Gen1 × 1, DPoC USB3.2 Gen1 × 1
UART	× 4
SDIO	× 2
ADC	× 2, universal ADC interface
I2C	× 8, for TP, Cameras, Sensors, etc.
SPI	× 6
PWM	× 4
Motor Driven	× 1
I2S	× 2
PCIe	× 1
KEYPAD	× 7,for pow,rst,etc.
GPIO	× 183
DMIC	× 2
GBE	× 1

AI	M328S
eDP	× 1
PCM	× 2